INPUTS (0.25° grid, 2977 cropland cells)

Climate
NEX-GDDP-CMIP6
5 GCMs, 3 periods

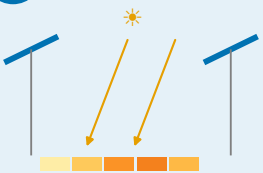
Soil
SoilGrids 2.0
profiles to 2 m

Cropland
MapSPAM area, irrigation
GGCMI sowing dates

Observed yields
ICRISAT state means
1995–2014

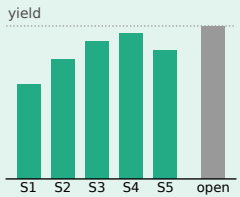
Terrain
Copernicus DEM
slope

1 Light below panels



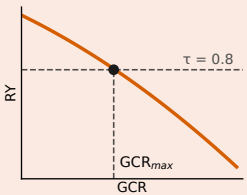
Hourly sun and sky light on 5 ground strips, for 3 layouts and GCR 0.05–0.7

2 Crop growth



DSSAT, 4 crops, irrigated and rainfed; N calibrated by state

3 Yield limit



Largest GCR with RY ≥ 0.66, 0.80 or 0.90

4 Potential



Crop area on slopes ≤ 5% × GCR → capacity and energy

Repeated for every cell × 5 climate models × 3 periods (1995–2014; 2041–2060 SSP2-4.5 and SSP5-8.5) → maps and totals as 5-model median and range